

Probabilistic and Uncertainty-Aware Deep Learning in Urban Air Quality (2023–2026)

Evaluated Research and Operational Systems

The following table presents eight verified items drawn from current research and operational deployments that satisfy the parameters of chemistry-transport hybrids, uncertainty-aware neural processes, deep ensembles, and European operational services. As requested, synthesis commentary has been strictly omitted. Where source data did not provide specific titles, authors, or verifiable DOIs, these fields have been explicitly marked as unavailable or unverified.

Target Question	Title & Authors	Venue & Year	DOI	Verbatim Abstract Sentence
Q1, Q2	OmniPM-Net: Bridging discrete and gridded PM10 forecasts via omni-query neural processes Authors: <i>Unavailable in source data</i>	<i>arXiv</i> 2026	UNVERIFIED	"Here we present OmniPM-Net, a Convolutional Conditional Neural Process (ConvCNP)-based fusion model that reconciles these two forecast types within a shared spatial representation." ¹
Q1, Q4	AEON-UP: Probabilistic Deep Learning for Urban Air	Pre-operational European Project 2026 (Launch)	UNVERIFIED	"In the AEON-UP project, we develop a transferable AI

	Quality Authors: Dr. Martin Ramacher, Dr. Matthias Karl (Helmholtz-Zentrum Hereon, Helmholtz Munich, RIFS Potsdam)			system for high-resolution urban air quality prediction across European cities and combine: physics-based chemistry transport models (CTMs) with probabilistic deep learning methods, in particular neural processes."²
Q1, Q4	Forecasting system for Urban Air Quality with automatic correction and web service for public dissemination Authors: M. Karl, S. Acksen, R. Chaudhary, M.O.P. Ramacher	<i>International Journal of Digital Earth</i> 2024	10.1080/17538947.2024.2359569	"It utilizes a unique bias correction algorithm that automatically adjusts boundary conditions and traffic emissions based on the observations recorded over the previous days."⁵
Q2	<i>Title Unavailable in Source Data</i> Authors:	<i>Scientific Reports</i> 2025	10.1038/s41598-025-05958-2	"Continuous-time neural processes, defined by differential

	Researchers at Research Center for Information Technology Innovation, Academia Sinica			equations, are limited by numerical solvers, restricting scalability and hindering the modeling of complex phenomena like neural dynamics-ideally addressed via closed-form solutions."⁷
Q2	AirRadar: A Deep Neural Network for Air Quality Inference Authors: <i>Unavailable in source data</i>	<i>arXiv</i> 2025	UNVERIFIED	"To address this limitation, we introduce AirRadar, a deep neural network designed to accurately infer real-time air quality in locations lacking monitoring stations by utilizing data from existing ones."⁸
Q2	Ensemble Deep Learning for Classification of Pollution Peaks	<i>WIT Transactions on Ecology and the Environment</i> 2023	UNVERIFIED	"The classification performance of ensemble decision tree (gradient boosting

	Authors: <i>Unavailable in source data</i>			machine (GBM)) models and ensemble deep learning (EDL) models are compared." ⁹
Q3	Assessing the Feasibility of Landsat-Driven NO₂ Prediction: A Spatial Cross-Validation Framework Authors: Amir Tahooni, Ata Kakroodi, Majid Kiavarz, Hossein Mansourian	ISPRS Annals 2026	UNVERIFIED	"LOSO CV revealed that while models achieved a pooled RMSE of $\approx 45.5 \mu\text{g}/\text{m}^3$, their mean R^2 was negative, indicating predictions at unseen locations were often worse than using the simple global mean." ¹⁰
Q3	Title <i>Unavailable in Source Data</i> Authors: <i>Unavailable in source data</i>	Atmosphere (MDPI) 2026	UNVERIFIED	"Using leave-one-station-out spatial cross-validation, we derived pollutant-specific uncertainty growth rates (α) from 3.84% to 5.62% per km, characterizing how calibration uncertainty increases with distance from reference

				stations (statistically significant for PM10 and O3, $p < 0.05$). ¹¹
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Direct Evaluation of Leave-One-Station-Out Cross-Validation (Q3)

Leave-one-station-out cross-validation (LOSOCV) is not universally treated as a routine or default standard practice across the entirety of the broad air quality machine learning literature, where random k -fold cross-validation or temporal hold-out strategies remain highly prevalent. However, within the specific sub-disciplines of spatial modeling, geographic information science, and hybrid sensor network calibration, LOSOCV is now explicitly recognized and demanded as a methodological necessity for evaluating spatial generalizability¹⁰.

If proposed as the primary validation design for a new model, it will not read as unusual or eccentric to domain experts. Instead, it will register as a deliberately strict and highly rigorous choice designed to expose the spatial overfitting that traditional cross-validation masks¹⁰. Studies utilizing LOSOCV emphasize that traditional validation yields optimistic performance

metrics (e.g., falsely inflating R^2 values), while spatial hold-outs like LOSOCV accurately reflect the often-severe performance degradation that occurs when predicting at unmonitored coordinates¹⁰. Consequently, implementing LOSOCV signals methodological maturity and a direct confrontation with the field's core challenge of spatial data sparsity¹⁰.

Works cited

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